

Integrating Human Factors into Insider Threat Detection

What is Insider Threat?

An insider threat is when a trusted person inside an organization causes harm.

Examples:

Employee stealing company data

Staff accidentally sending confidential file

A) Intentional Insider Threat (Malicious)

Person purposely causes harm.

Examples:

Selling confidential data

B) Unintentional Insider Threat (Accidental)

Person does not mean harm but causes it by mistake.

Examples:

Weak passwords

Problem Statement

Current detection systems mainly focus on:

- * System logs
- * File access patterns
- * Network activity

They ignore:

- * Human emotions
- * Stress
- * Work pressure
- * Lack of awareness
- * Organizational issues

Research Gap:

Human factors are underused in ML-based detection.

Why This Research Matters (Important Statistics)

74% of breaches involve human error
43% are non-malicious mistakes

Q. Why this percentage does not tally to 100?

74% means most security breaches happen because of human mistakes.

43% does NOT mean 43 out of 100 total breaches.
It means 43% of the 74 human-error breaches are accidental.

This equals about 32 accidental breaches out of 100 total breaches.

This shows many insider threats happen unintentionally because of stress, lack of awareness, or poor training.

That is why technical monitoring alone is not enough, and human factors must be included in detection systems.

Security Breach Statistics

74% of all security breaches involve human error

Among these human-error breaches:

43% are accidental (non-malicious)

LITERATURE REVIEW

Existing studies focus on:

- Technical anomaly detection
- ML/DL methods
- Synthetic datasets

Limitations:

- Weak human-factor analysis
- Unrealistic datasets
- Less accidental threat focus

Methodology (PRISMA)

Methodology means the process or steps used to conduct the research. It explains how the study was carried out, how data or papers were selected, and what methods were used.

PRISMA stands for:

Preferred Reporting Items for Systematic Reviews and Meta-Analyses

It is a structured method or framework used in research papers to collect, filter, and select research studies in a proper and transparent way.

Why is PRISMA Used?

- Select only relevant research papers
- Remove duplicate or unrelated studies
- Make the research process transparent
- Increase reliability and accuracy
- Show how papers were chosen

How PRISMA Works (4 Main Steps)

1. Identification

Researchers search papers from databases.

Example:

- IEEE
- ACM
- Springer
- ScienceDirect
- Google Scholar

Goal:

Find all related papers

2. Screening

Unrelated and duplicate papers are removed.

Example:

Remove: unrelated cybersecurity topics, duplicate papers

Goal:

Keep only useful papers

3. Eligibility

Researchers read full papers carefully.

Check:

Is it related to insider threats?

Is it relevant to human factors?

Is it recent and useful?

Goal:

Choose high-quality papers

4. Included Studies

Final selected papers are included in the review.

Check:

Is related to the research topic

Matches the study objective

Contains useful and relevant data

Goal:

Only best and relevant studies remain

Human Factors Taxonomy

- Human factors means the behaviour, actions, emotions, mistakes, habits, and decisions of people that can affect a system, workplace, or security.
- In cybersecurity (especially insider threats), human factors focus on how a person's behavior can cause security problems, either intentionally or unintentionally.
- Taxonomy means classification or grouping things into categories based on similarities.
- It is a method used to organize and classify things properly so they are easier to understand and study.
- They grouped human factors into:

A) Individual Factors

B) Motivational Factors

C) Organizational Factors

D) Situational / Task-related Factors

A) Individual Factors

Personal traits and cognitive limitations that influence risky or careless behaviour.

Examples:

- Personal traits
- lack of knowledge
- Fear

B) Motivational Factors

Reasons that make a person intentionally misuse access, such as money, revenge, anger, or personal benefit.

Examples:

- Financial reward
- Revenge
- Addiction
- Personal gain

C) Organizational Factors

Workplace policies, leadership, and culture that shape insider threat risks.

Examples:

- Poor policies
- Weak leadership
- Bad communication

D) Situational / Task-related Factors

Work pressure and task complexity that increase accidental security mistakes.

Examples

- Complex tasks
- Stress
- Fatigue

ML/DL model DetectionTechniques

ML Models

1. SVM (Support Vector Machine)

Purpose:

Separates normal and suspicious behaviour using a boundary line.

Good:

Works well for small structured data.

Bad:

Not suitable for very large complex data.

2. Decision Tree

Purpose:

Makes decisions using yes/no conditions.

Good:

Easy to understand and explain.

Bad:

Can overfit and make wrong predictions.

3. Random Forest

Purpose:

Uses many decision trees together to improve prediction.

Good:

More accurate than one decision tree.

Bad:

Needs more processing power.

4. KNN (K-Nearest Neighbors)

Purpose:

Compares current behaviour with past similar behaviours.

Good:

Simple and effective.

Bad:

Slow for large datasets.

Deep Learning Models

1. GRU (Gated Recurrent Unit)

Meaning:

A type of Deep Learning model used to analyze data that happens over time.

Why used?

To understand employee behavior patterns over time.

Example:

If an employee suddenly starts downloading many files at midnight, GRU can detect unusual behavior.

2. LSTM (Long Short-Term Memory)

Meaning:

A special type of neural network that remembers past information for a long time.

Why used?

Used when past employee actions matter.

Example:

If a user slowly behaves suspiciously over weeks, LSTM remembers previous activities.

3. CNN (Convolutional Neural Network)

Meaning:

A Deep Learning model mainly used for image recognition but also pattern detection.

Why used?

To identify hidden patterns in logs and data.

Example:

Detecting suspicious activity patterns in system logs.

4. RNN (Recurrent Neural Network)

Meaning

A neural network designed for sequential data.

Why used?

Used when data changes over time.

Example:

Tracking login activity sequence of employees.

5. GNN (Graph Neural Network)

Meaning

A model that studies relationships between connected things.

Why used?

To study employee relationships and network behavior.

Example

Checking who accessed which system or shared files.

6.FFNN (Feed Forward Neural Network)

FFNN is a simple neural network where data moves in one direction (input to output) and is used for pattern recognition, prediction, and classification.

Why used?

Basic prediction and classification.

Example

Predict whether employee behavior is safe or risky.

7. DBN (Deep Belief Network)

Meaning

A deep learning model used to learn hidden patterns.

Why used?

Used for anomaly detection.

Example

Detecting abnormal employee behavior.

8. ELK (Elasticsearch – Logstash – Kibana)

Meaning

A toolset used for log monitoring and data analysis.

Why used?

To monitor employee system activities.

Components:

Elasticsearch

→ Stores and searches data

Logstash

→ Collects logs

Kibana

→ Creates visual dashboards

Example

Monitoring login failures or suspicious access.

Datasets Used

What is a Dataset?

A collection of user activity data used to train and test insider threat detection models

What is Synthetic Data?

Artificially generated data that simulates real employee behaviour for research purposes.

Common Datasets Used

1. CERT Dataset

Most widely used benchmark dataset

Includes login, email, file, web, and device activity

Limitation: Synthetic data

2.Enron Email Dataset:
Real employee emails

Limitation: Only email data

3.TWOS Dataset:
Workplace activity logs

Limitation: Small size

4.RTU Dataset:
User behaviour logs

Limitation: Limited realism.

Limitations in Existing Research

- Heavy Dependence on Synthetic Data
Models rely on artificial datasets that may not reflect real behaviour.
- Lack of Human Factor Integration
Most systems ignore stress, motivation, awareness, and workload.
- Poor Real-World Validation
Models are rarely tested in actual enterprise environments.
- Data Imbalance Problem
Insider attacks are rare, making detection difficult.
- Limited Explainability
Deep learning models often act as black boxes.

Future Research Direction

- Build Richer Datasets
Combine technical logs with human behavioural data.
- Hybrid Detection Models
Integrate ML, DL, and human factor analysis.
- Focus on Unintentional Threats
Detect accidental insider risks caused by human error.
- Explainable AI (XAI)
Provide clear reasons for threat detection decisions.
- Real-World Testing
Validate models in practical organizational settings.

Conclusion

- Most studies focus on technical behaviour only
- Human factors are often ignored
- Synthetic datasets limit accuracy

Thank You